```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/A2_part_4$ ./p_integrate 4
Sequential result: 3.1415926536
Sequential time: 2.884971 seconds
Result of numerical integration: 3.1415926536
Threads: 4
Execution time: 0.867826 seconds
```

```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/A2_part_4$ ./main_merge 4
Sequential Merge Sort Time: 8.587 ms
Parallel Merge Sort Time: 3.467 ms
Recursive Merge Sort Time: 3.326 ms
Thread Pool Merge Sort Time: 3.952 ms
All sorting approaches match: YES
```

```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/A2_part_4$ ./p_bubble_1 4
Sequential Bubble Time: 10573.402 ms (sorted: YES)
Parallel Bubble_1 Time (4 threads): 8817.298 ms (sorted: YES)
```

```
keatheat@DESKTOP-4MCQ7N9:~/Parallel-Computing/A2_part_4$ ./p_bubble_2 4
Sequential Bubble Time: 10564.674 ms (sorted: YES)
Parallel Bubble_2 Time (4 threads): 7231.432 ms (sorted: YES)
```